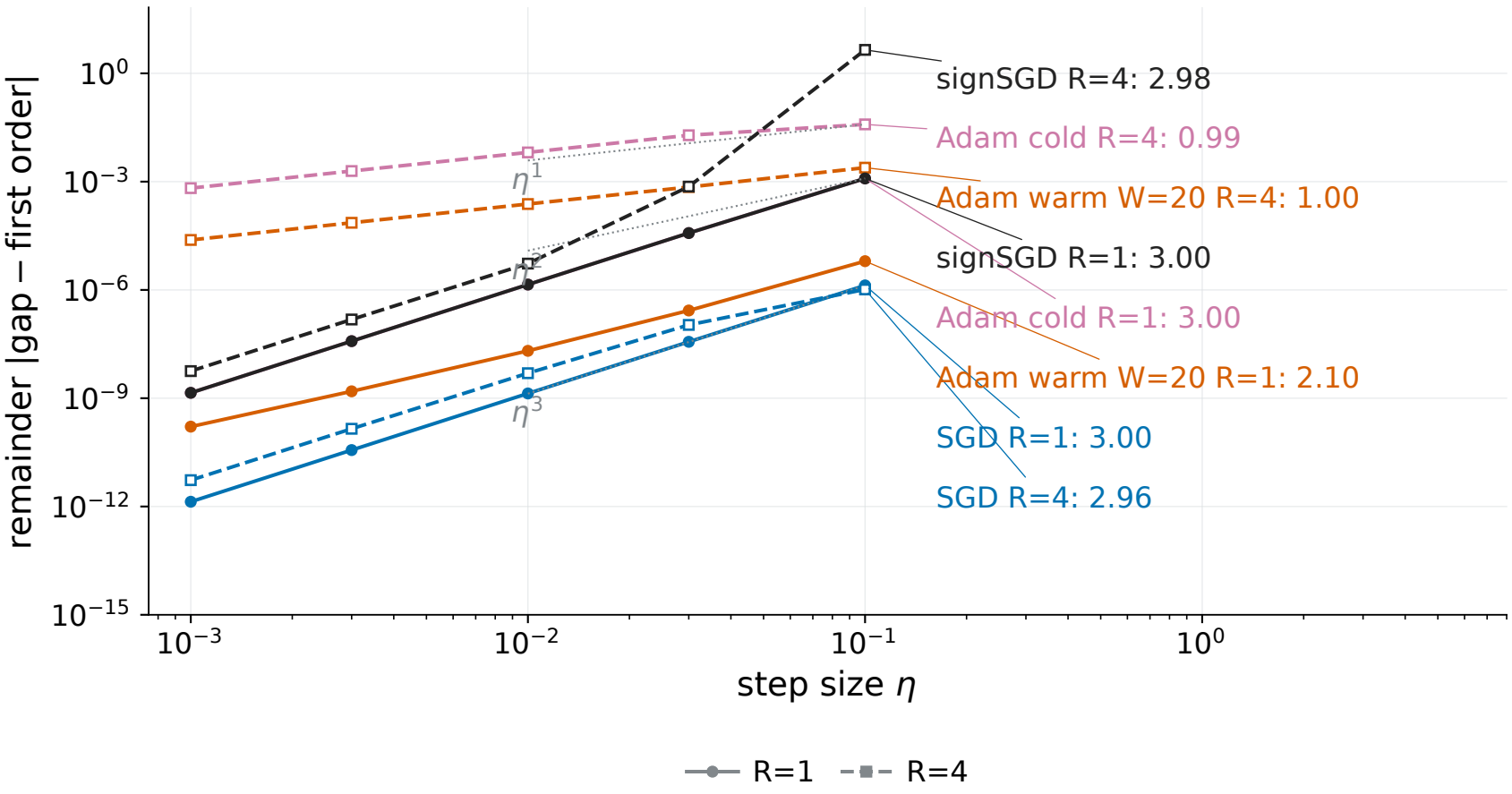
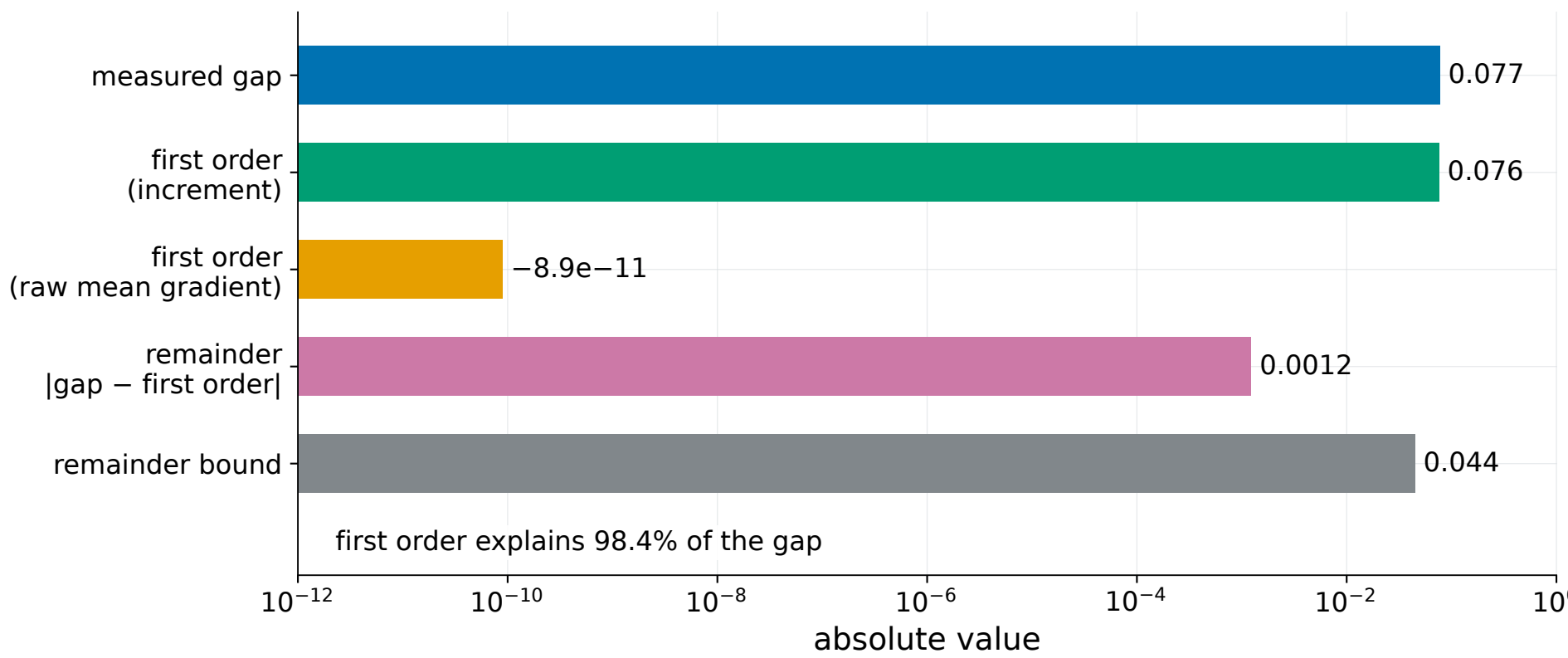
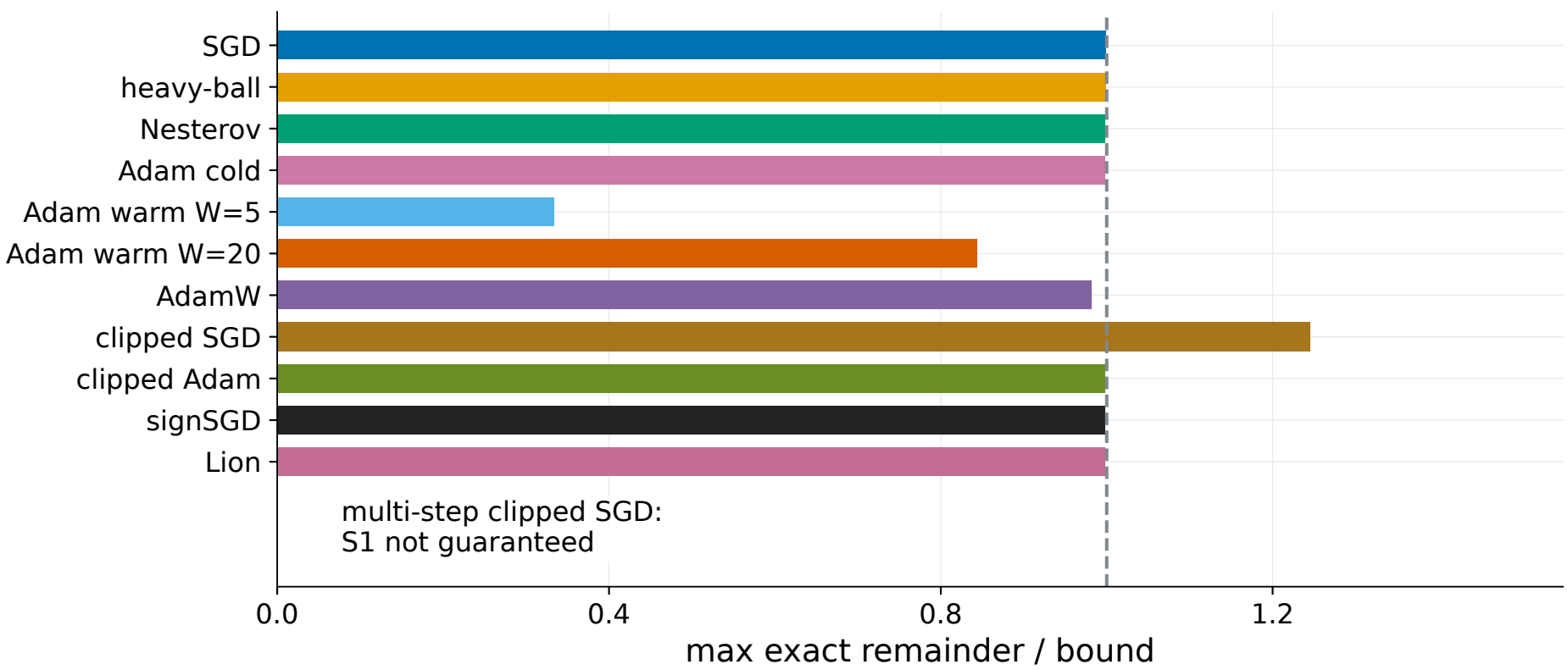
(a) Remainder order in η
(moment N=3)



(b) Cold-Adam gap:
increment vs raw gradient



(c) Bound check in exact cells
(R = 1, 4)



R=40 Monte Carlo: 2 of 330 cells
above bound after 4 SE (sign-type)

Controlled small network; not a real-model result. Bounds use empirical curvature estimates; S1 is not certified.